

Nithin E

Data Scientist

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Summary

I am an AI & Data Science graduate who loves turning complex problems into working applications. Through building projects like an autonomous cybersecurity agent and a brain tumor detection model, I've gained solid, hands-on experience with Python, machine learning frameworks, and databases. My focus is on leveraging these tools to build robust, end-to-end AI solutions that deliver real-world impact.

Technical Skills

- **Programming & Analytics:** Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Keras), SQL
- **AI/ML Frameworks:** TensorFlow, Keras, PyTorch, Scikit-learn, LangGraph, LLM APIs (Groq Llama, Mistral), GNN, Isolation Forest
- **Database Systems:** MySQL, MongoDB
- **Tools & Platforms:** Jupyter Notebook, VS Code, Git/GitHub, Power BI Service

Projects

AI-Powered Cyber Defense System (Project Defend)

GitHub

React, Node.js, Python, GNN, PyTorch, LangGraph, LLMs (Groq Llama, Mistral)

- Developed a MERN-based E-learning platform integrated with an autonomous AI defense agent for real-time threat mitigation.
- Engineered a multi-model anomaly detection system using **GNNs**, **Isolation Forests**, and OC-SVM to dynamically block DoS, XSS, and SQLi attacks.
- Designed explainable agentic workflows with **LangGraph** and LLMs to automate IP firewalling and generate live security telemetry.

Brain Tumor Detection Model

GitHub

Python, Scikit-learn, TensorFlow, Keras, Pandas

- Developed a machine learning pipeline to classify tumors as malignant or benign based on cellular features, achieving **92.00% accuracy**.
- Engineered robust data processing workflows for feature extraction, analyzing nucleus size, texture, perimeter, and compactness.
- Trained and evaluated **Random Forest** and Deep Learning models to optimize diagnostic performance for medical AI research.

PacTune — Desktop Music Player

GitHub

Rust, GTK4, Libadwaita, GStreamer, Relm4

- Built a high-performance, native Linux desktop music player in **Rust** using **GTK4** and Libadwaita for seamless GNOME integration.
- Integrated **GStreamer** for multi-format audio playback and engineered an automated, auto-syncing lyrics fetcher (.lrc support).
- Designed a reactive UI with Relm4, featuring album artwork grids, media key support, and automated installation scripts.

Certifications

- **Data Analyst Course — PYSpiders (2024):** Python, Power BI, and SQL.
- **Data Science Course — Udemy (2024):** ML, Statistics, Python, and Power BI with real-world projects.

Education

B.Tech in Artificial Intelligence & Data Science | CGPA: 7.2/10

2026

Mahendra Engineering College, Tamil Nadu

Relevant Coursework: Probability & Statistics, Machine Learning, Deep Learning, Data Structures